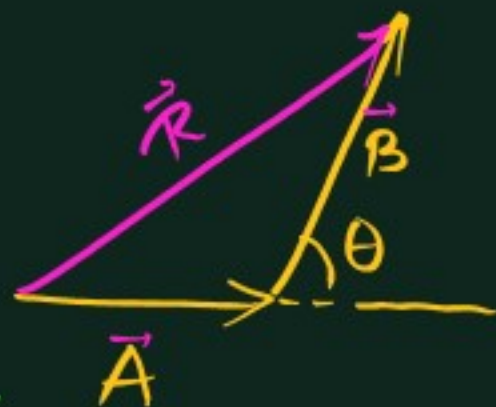


Vectors.

① Addition



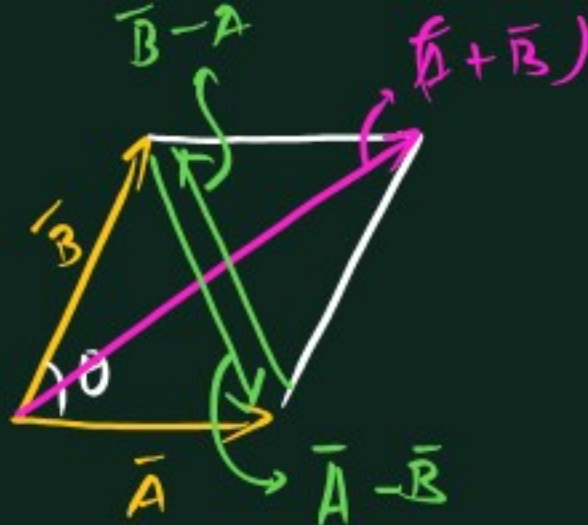
$$R = \sqrt{A^2 + B^2 + 2AB \cos \theta}$$

$$R_{\max} = A + B \quad (\theta = 0^\circ)$$

$$R_{\min} = A - B \quad (\theta = 180^\circ)$$

($\cos 180^\circ = -1$)

②

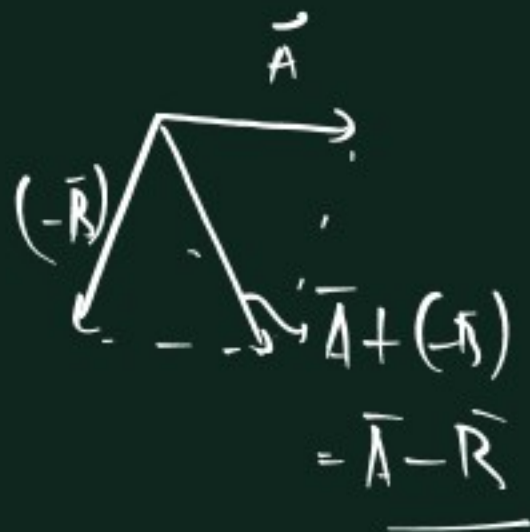


③ Subtraction

$$|\vec{A} - \vec{B}| = \sqrt{A^2 + B^2 - 2AB \cos \theta}$$

$$\vec{A} + (-\vec{B})$$

(Between $\vec{A}$ & $\vec{B}$)



Q



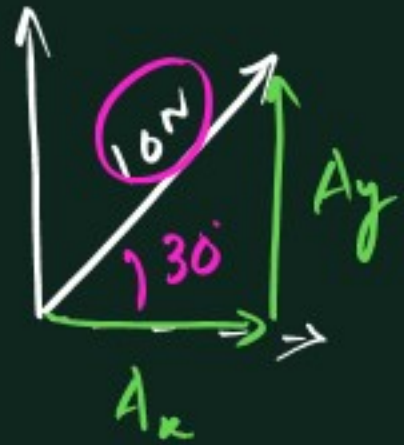
$$\underline{Ans:} \quad R = \sqrt{3^2 + 5^2 + 2 \times 3 \times 5 \times \frac{1}{2}}$$

$$= \sqrt{49} = \underline{7N}$$

$$2N \leq |\vec{A} + \vec{B}| \leq 8N$$



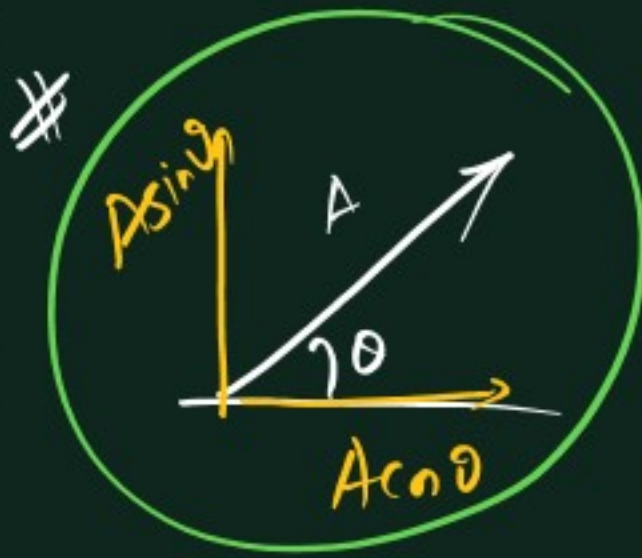
④ Components



$$\sin 30 = \frac{A_y}{10}$$

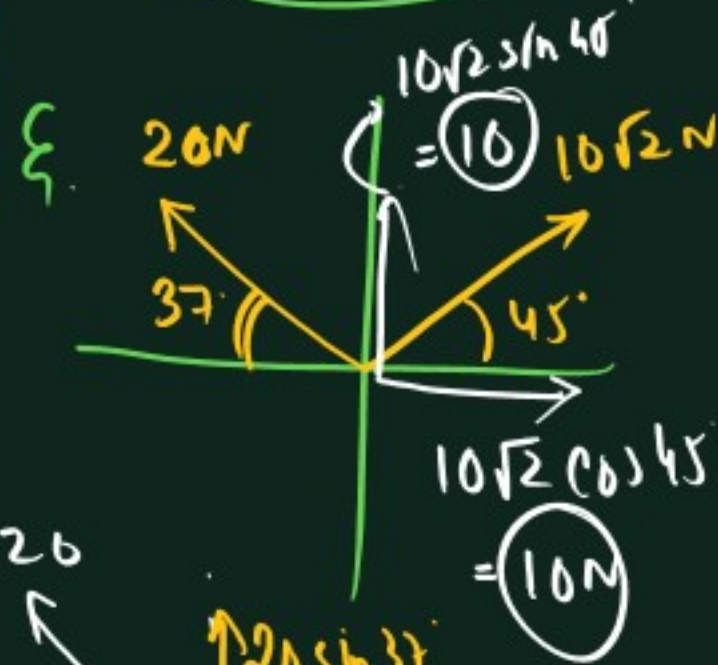
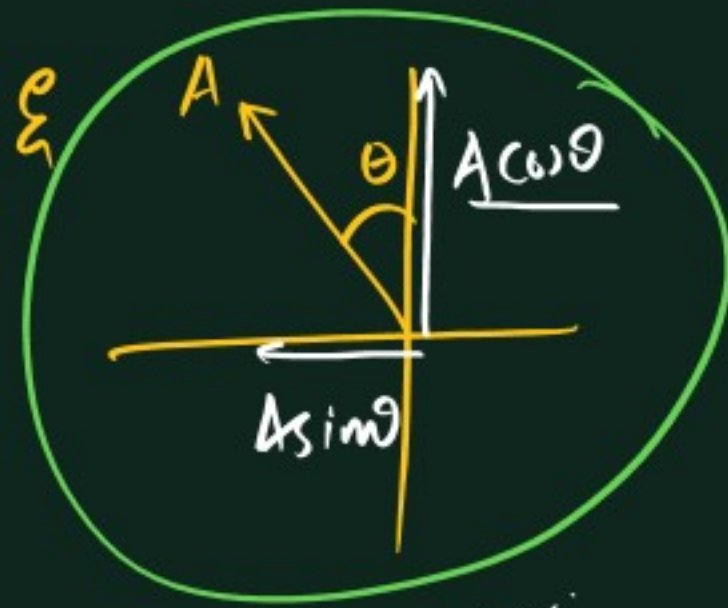
$$\Rightarrow A_y = 10 \sin 30$$

$$A_x = 10 \cos 30$$



$$\vec{A} \rightarrow y\text{-comp} = 0$$

$$\vec{A} \uparrow \Rightarrow x\text{-comp} = 0$$

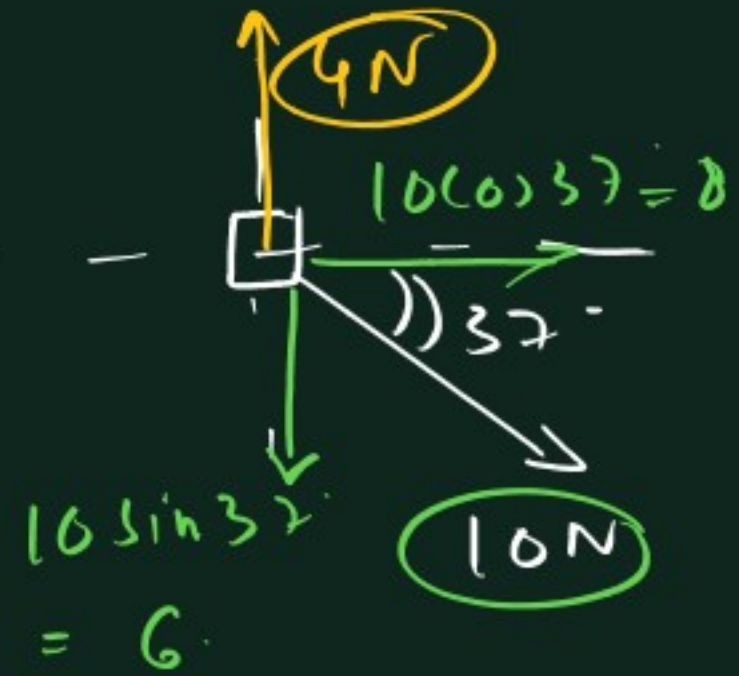


$$20 \sin 37 = 12 \text{ N}$$

$$20 \cos 37 = 20 \times \frac{4}{5} = 16 \text{ N}$$

$$\cos 37 = \frac{4}{5} \\ \sin 37 = \frac{3}{5}$$

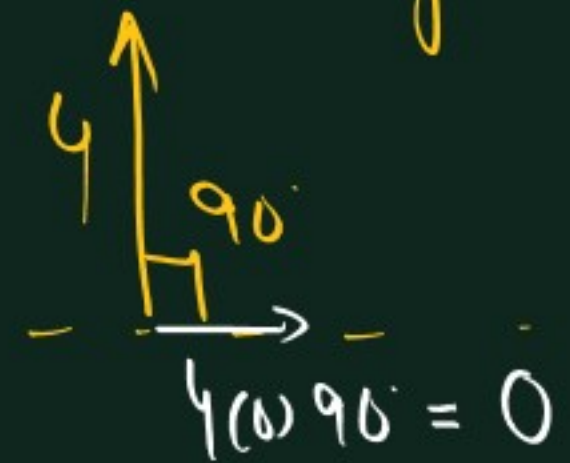
Ex.



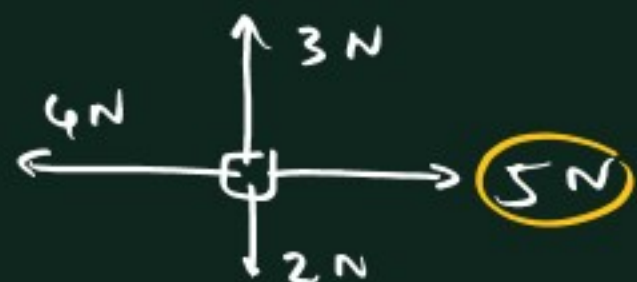
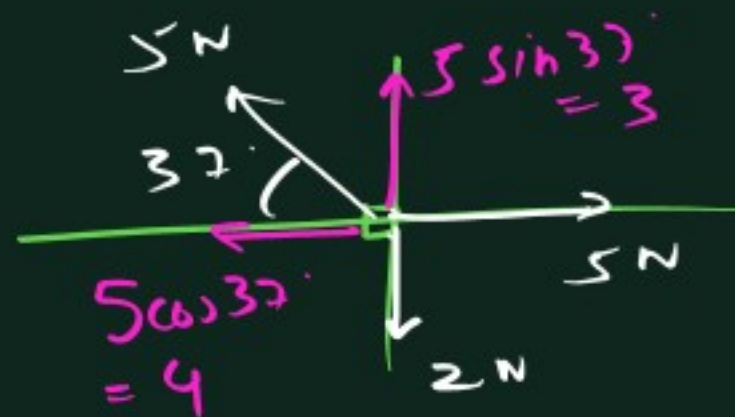
4N

$$\rightarrow x\text{-comp} = 0$$

$$\rightarrow y\text{-comp} = 0$$

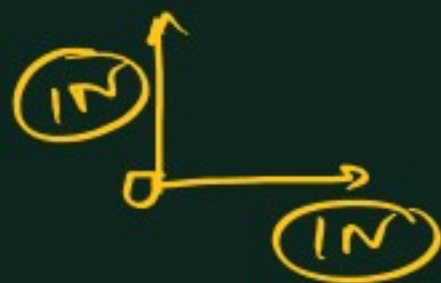


ξ



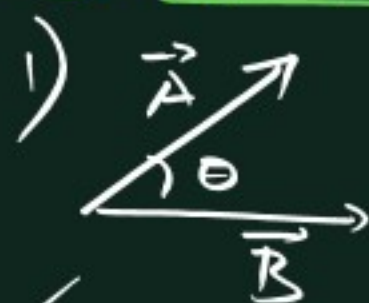
$$\textcircled{x} \Rightarrow 5 + (-4) = \textcircled{1N}$$

$$\textcircled{y} \Rightarrow 3 + (-2) = \textcircled{1N}$$



$$\vec{F}_{\text{net}} = 1\hat{i} + 1\hat{j}$$

⑤ Dot Product



$$\vec{A} \cdot \vec{B} = AB \cos \theta$$



$$\cos 120 = -\frac{1}{2}$$

$$\vec{A} \cdot \vec{B} = 6 \times 2 \times \left(-\frac{1}{2}\right) = \textcircled{-6}$$

$$\vec{A} \cdot \vec{B} = 0 \text{ if } \theta = 90^\circ$$

$$\begin{cases} \vec{A} = 2\hat{i} + 3\hat{j} \\ \vec{B} = 4\hat{i} - 5\hat{j} \end{cases}$$

$$\vec{A} \cdot \vec{B} = 2 \times 4 + 3 \times (-1) = \textcircled{3}$$

$$\textcircled{Q} \quad \textcircled{\hat{A}} = \underline{a\hat{i} + b\hat{j}}$$

45° with $(\hat{i} + \hat{j})$

60° with $(3\hat{i} - 4\hat{j})$

$$\cos 45^\circ = \frac{(a\hat{i} + b\hat{j}) \cdot (\hat{i} + \hat{j})}{|\hat{A}| |\hat{i} + \hat{j}|}$$

$$\Rightarrow \frac{1}{\sqrt{2}} = \frac{a + b}{1 \times \sqrt{2}}$$

$$\Rightarrow \underline{a + b = 1} \text{ --- } \textcircled{1}$$

$$\cos 60^\circ = \frac{(a\hat{i} + b\hat{j}) \cdot (3\hat{i} - 4\hat{j})}{1 \times \sqrt{3^2 + 4^2}}$$

$$\Rightarrow \frac{1}{2} = \frac{3a - 4b}{5}$$

$$\Rightarrow \underline{5 = 6a - 8b} \text{ --- } \textcircled{2}$$

Laws of motion

Force

→ Push / pull
→ change speed / direction /
shape - size

unit - N

$$[1 \text{ N} = 10^5 \text{ dynes}]$$

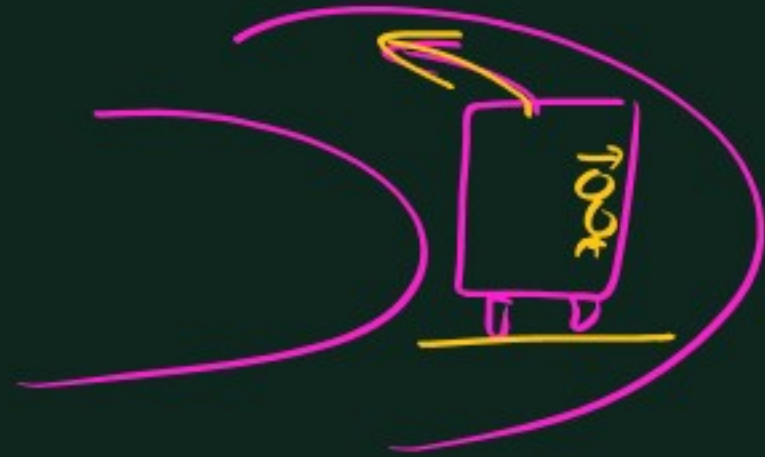
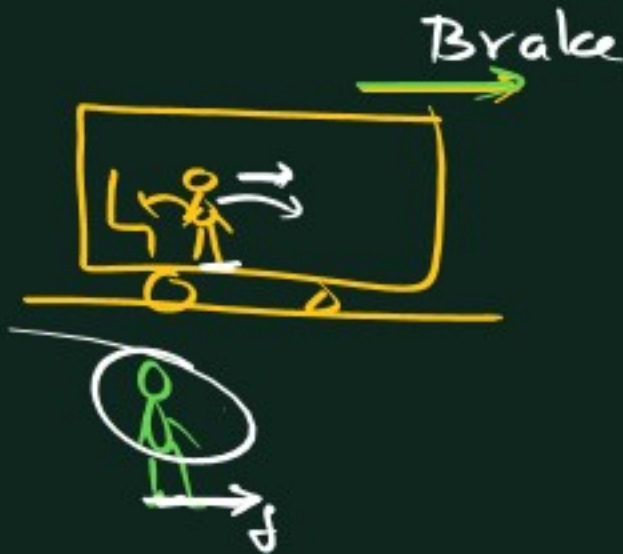
1st law of motion

Inertia of

1) Rest → Bus starts
falling back

2) Motion
Bus brakes &
fall forward

3) Direction → Turning of
bus

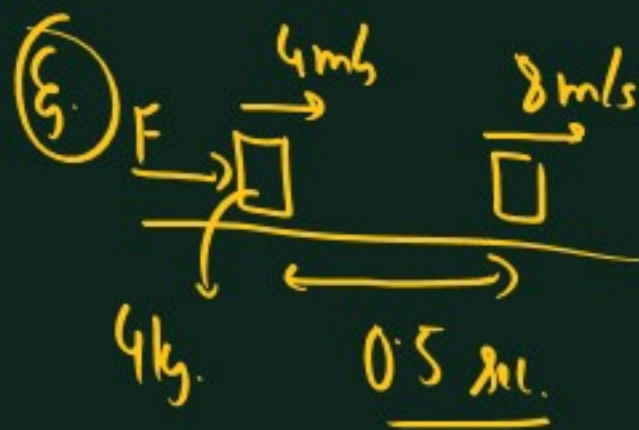


2nd law

✓ $F_{\text{net}} = \frac{\text{change in momentum}}{\text{time for the change}}$

$$\Rightarrow F_{\text{net}} = \frac{\Delta p}{\Delta t}$$

$p = mv$



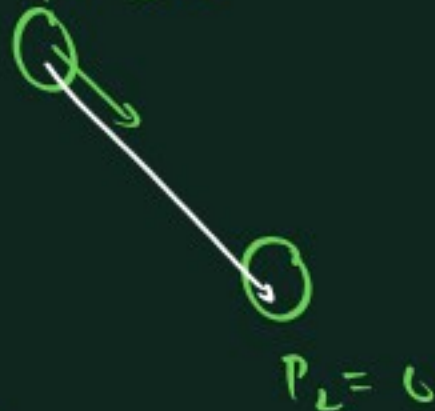
$$\begin{aligned} \text{A2: } F &= \frac{p_2 - p_1}{t} \\ &= \frac{4 \times 8 - 4 \times 4}{(1/2)} \\ &= \underline{\underline{32 \text{ N}}} \end{aligned}$$

$$\begin{aligned} F_{\text{net}} &= \frac{p_2 - p_1}{\Delta t} \\ &= \frac{mv - mu}{\Delta t} \\ \Rightarrow F_{\text{net}} &= m \left(\frac{v - u}{t} \right) \end{aligned}$$

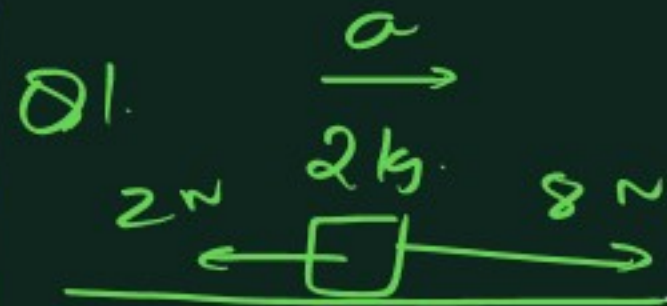
$$\begin{aligned} v &= u + at \\ \Rightarrow \frac{v - u}{t} &= a \end{aligned}$$

$$\boxed{F_{\text{net}} = ma}$$

$p_1 = 50$

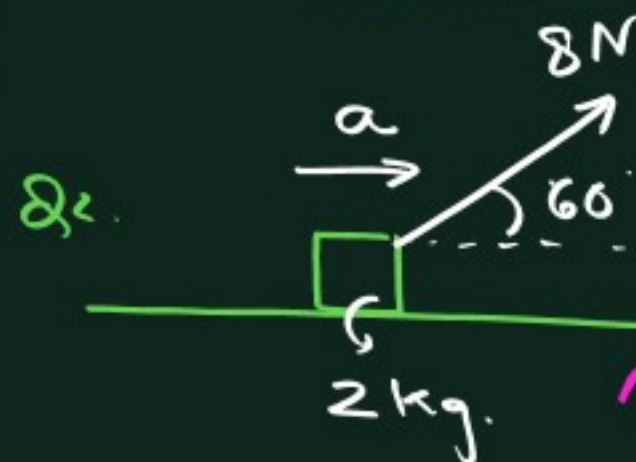


$$\begin{aligned} F &= \frac{0 - 50}{t} \\ \Rightarrow F &= -\frac{50}{t} \end{aligned}$$

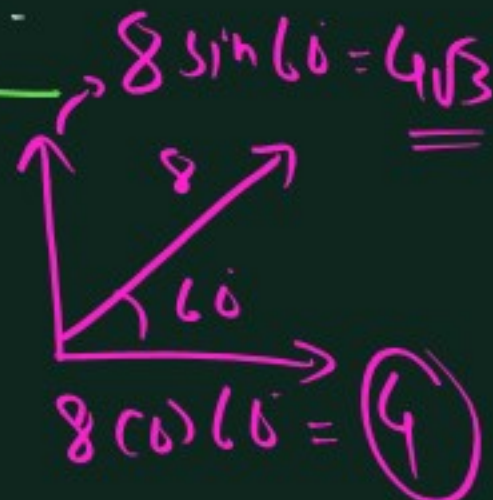


Find acceleration

$$\begin{aligned} \text{A2: } F_{\text{net}} &= ma \\ \Rightarrow 8 - 2 &= 2 \times a \\ \Rightarrow \boxed{a = 3 \text{ m/s}^2} \end{aligned}$$



$a = ??$



$$8 \sin 60 = 4\sqrt{3}$$



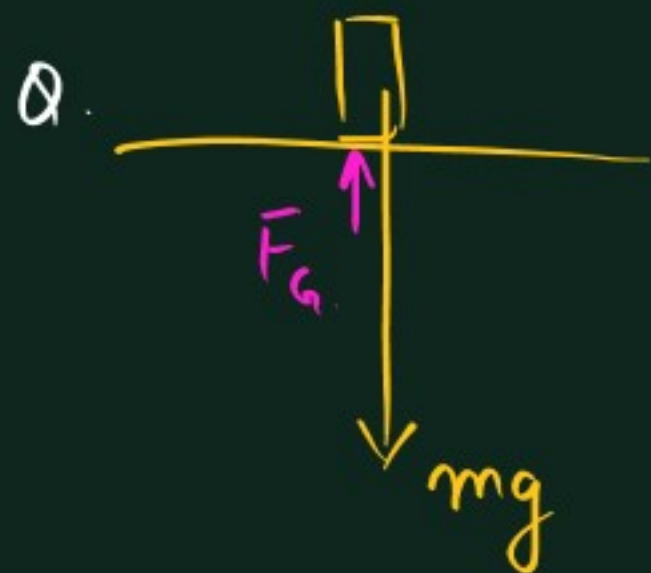
$$8 \cos 60 = 4$$



$$F = ma$$

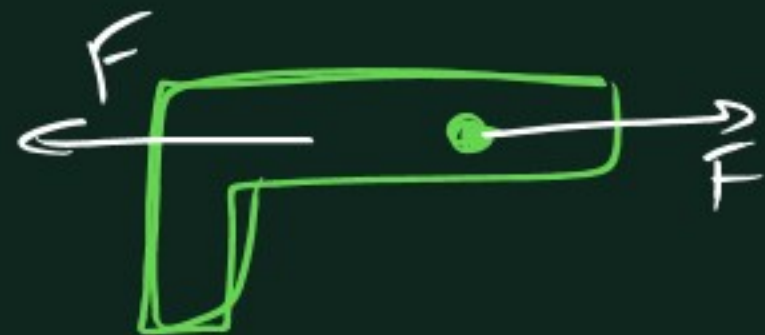
$$\Rightarrow 4 = 2 \times a$$

$$\Rightarrow \underline{a = 2 \text{ m/s}^2}$$



Action-Reaction pairs?

$$\left(\begin{array}{l} mg \longleftrightarrow mg \\ N \longleftrightarrow N \end{array} \right)$$



Forces

Contact

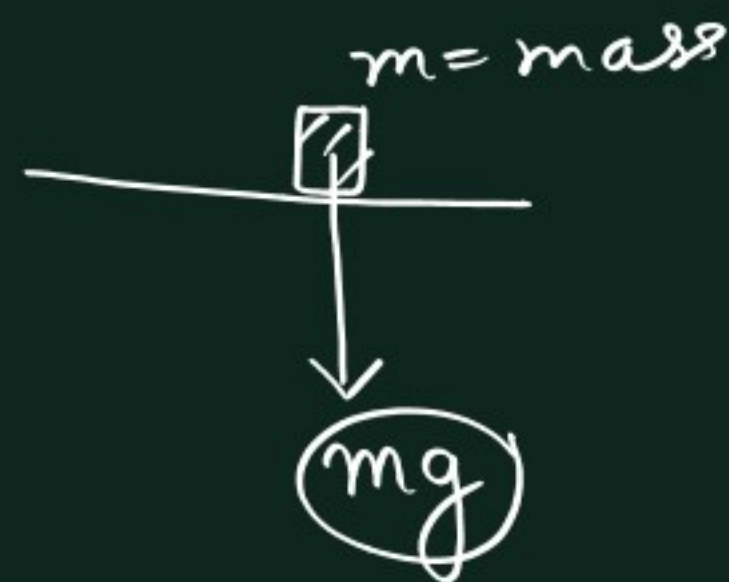
- friction
- Normal
- Tension
- Spring

Non-Contact (field force)

- Gravity
- magnetic
- electrostatic

→ gravitational / Electromagnetic (charges) / Strong nuclear force / Weak force

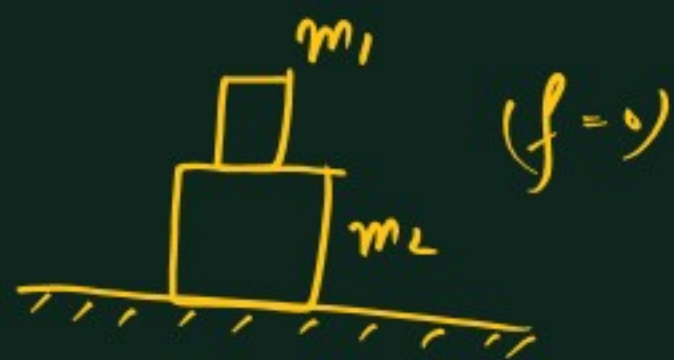
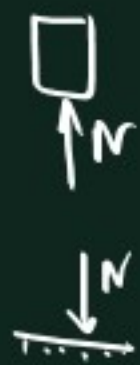
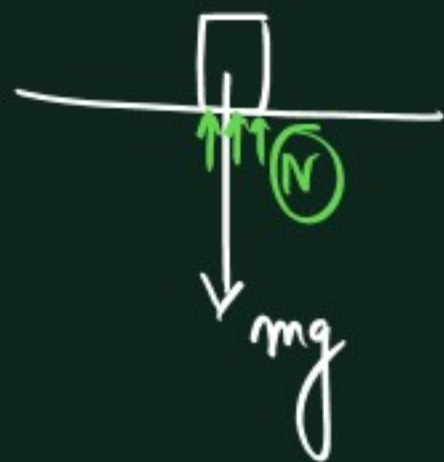
1) gravitational force



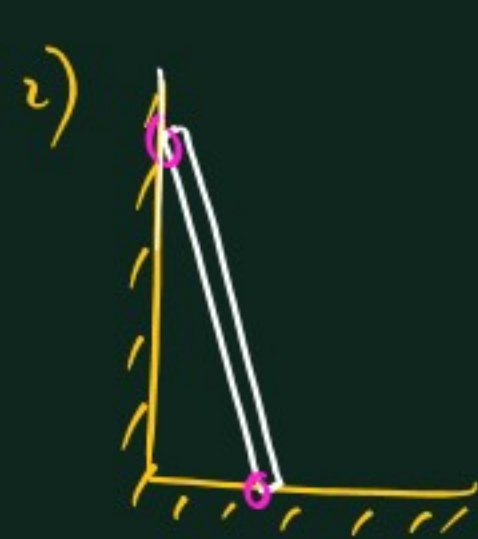
→ $g = 9.8 \text{ m/s}^2$
= acceleration due to gravity

→ Vertically down

2) Normal $\rightarrow$ (geometric) $\perp$

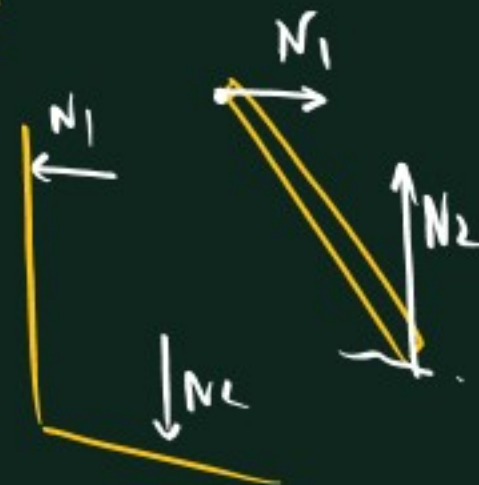


Draw all forces on m_1 & m_2

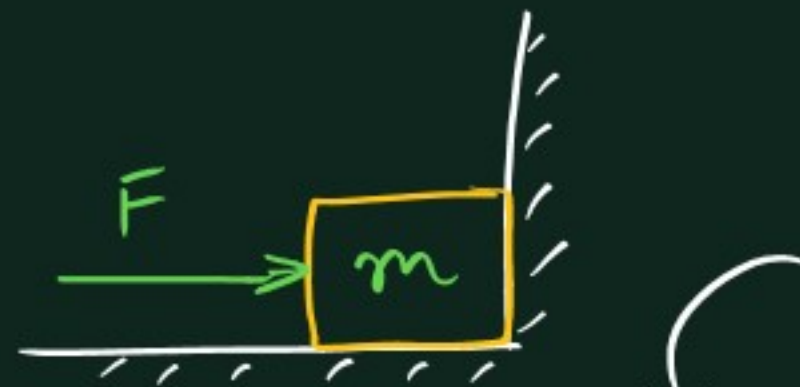


Draw N forces on ladder.

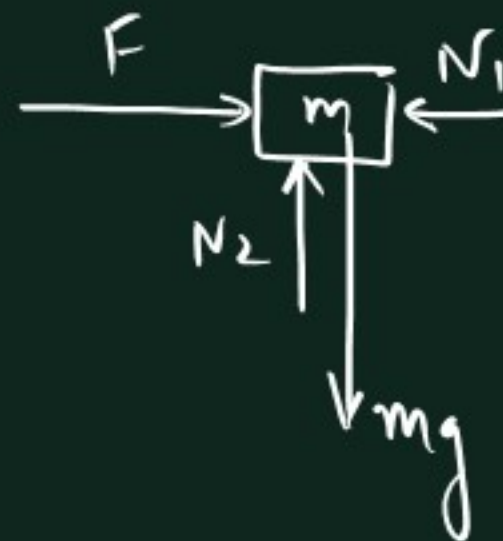
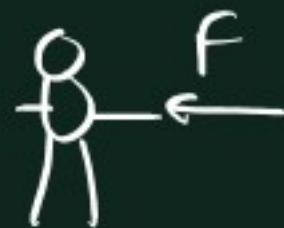
A-



3)



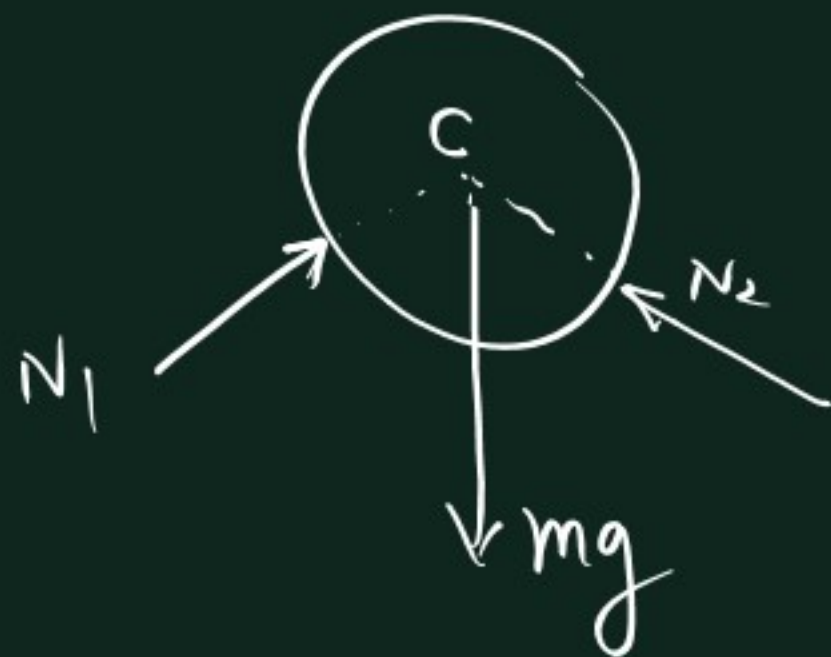
Draw forces on m

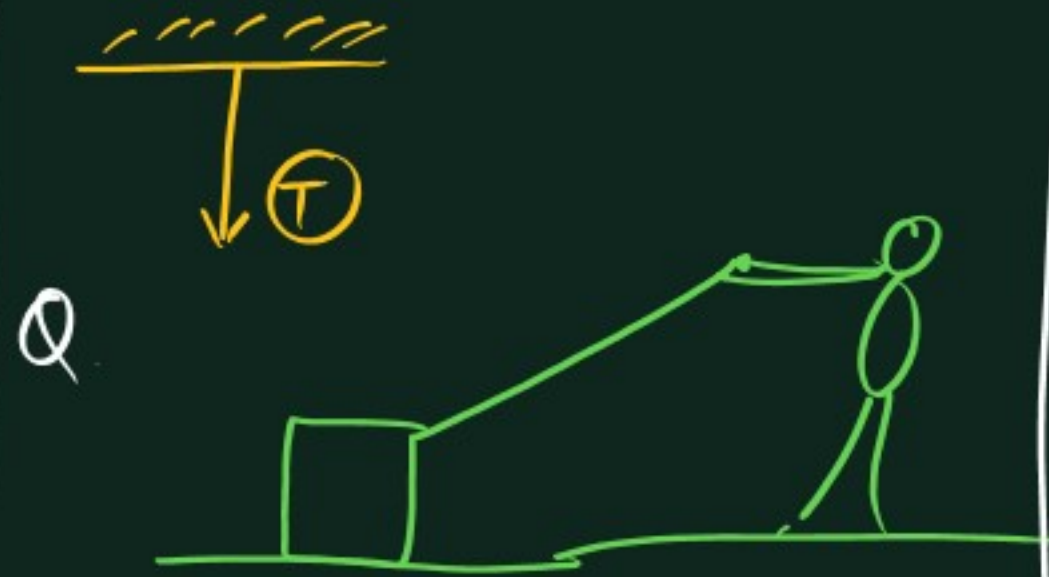
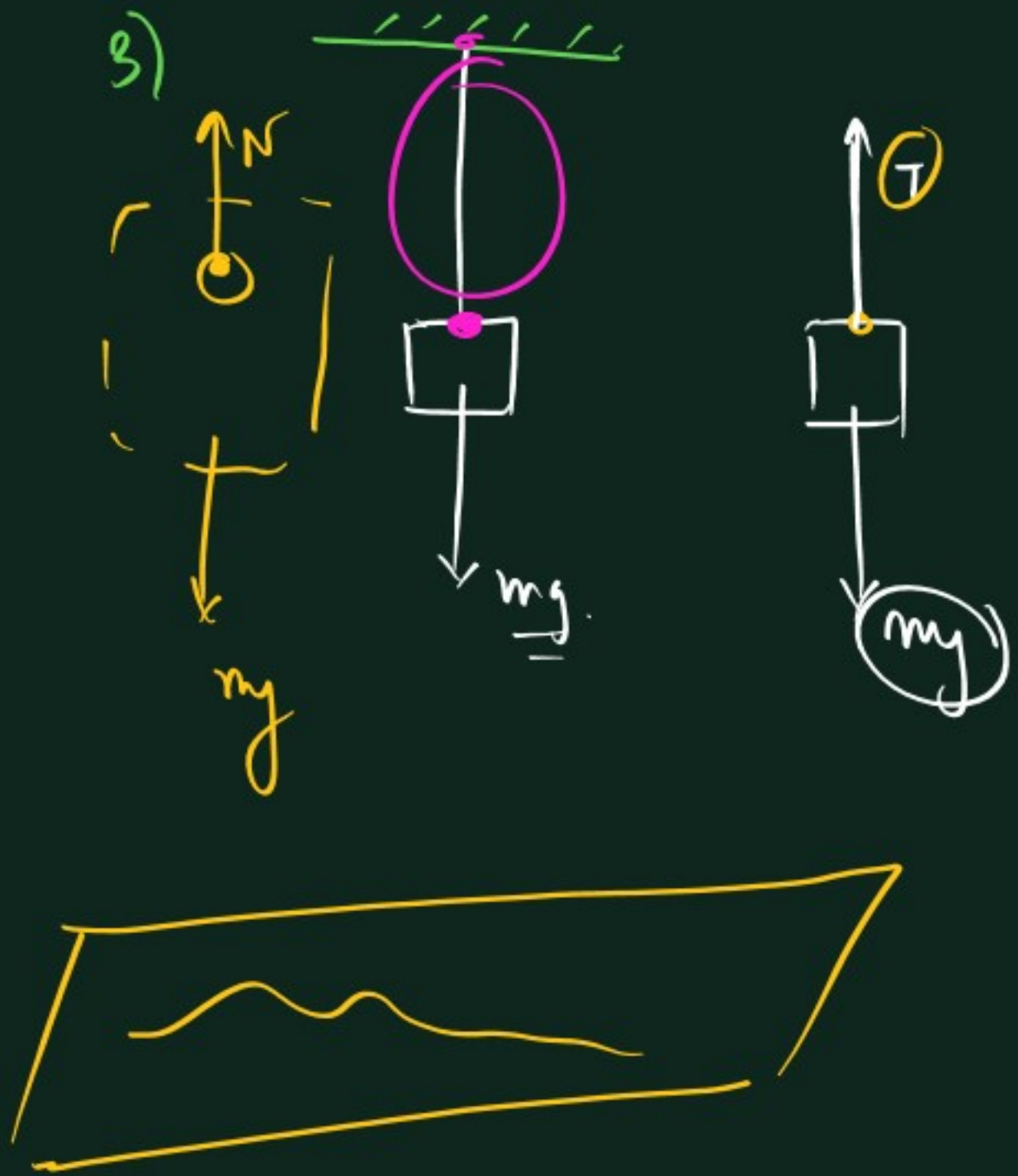


4)

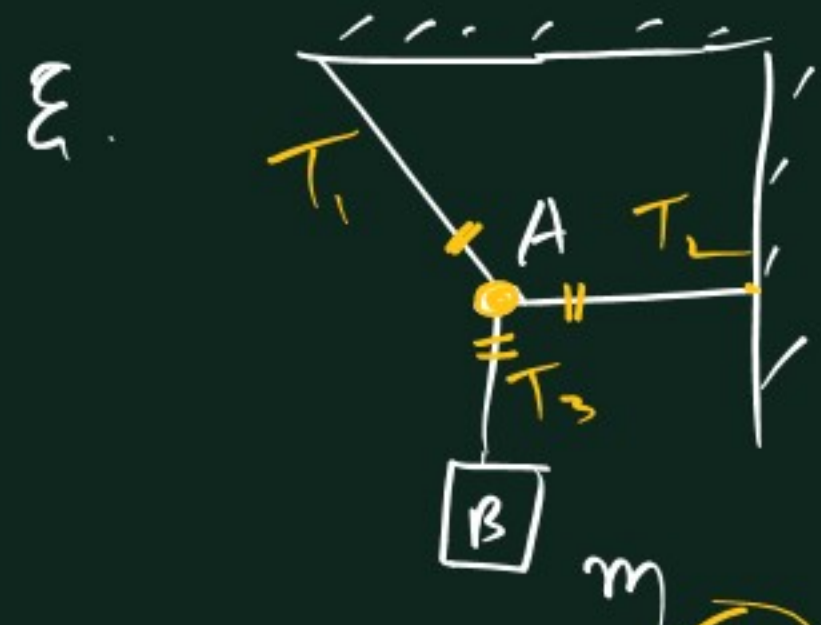


Draw forces on sphere

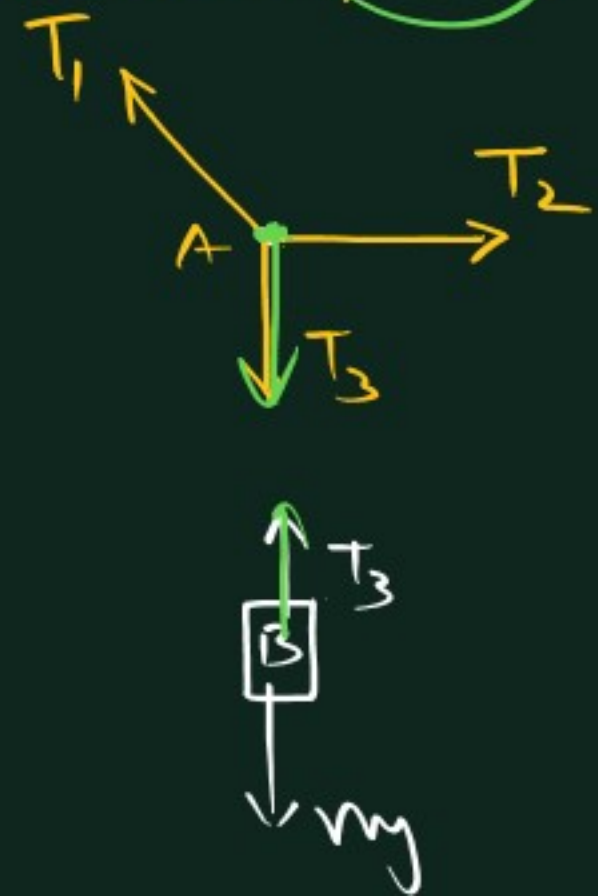




Draw forces on box.



- 1) Forces on box B
- 2) Forces on point A



$$\boxed{a=0} \text{ (rest, constant velocity)}$$

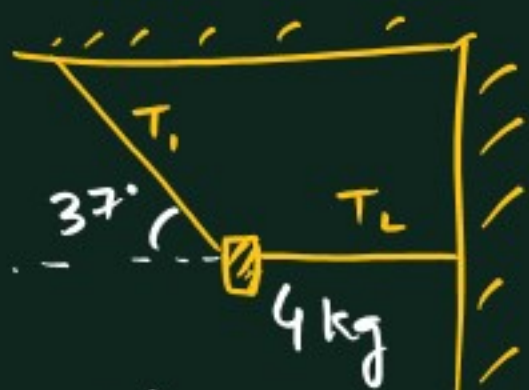
$$F_{\text{net}} = m\vec{a} = 0$$

$$\Rightarrow \boxed{F_{\text{net}} = 0}$$

FBD

$$\begin{array}{l} \text{Along } x, \quad \text{Net } F = 0 \quad \boxed{\sum F_x = 0} \\ \text{Along } y, \quad \text{Net } F = 0 \quad \boxed{\sum F_y = 0} \end{array}$$

Q1.

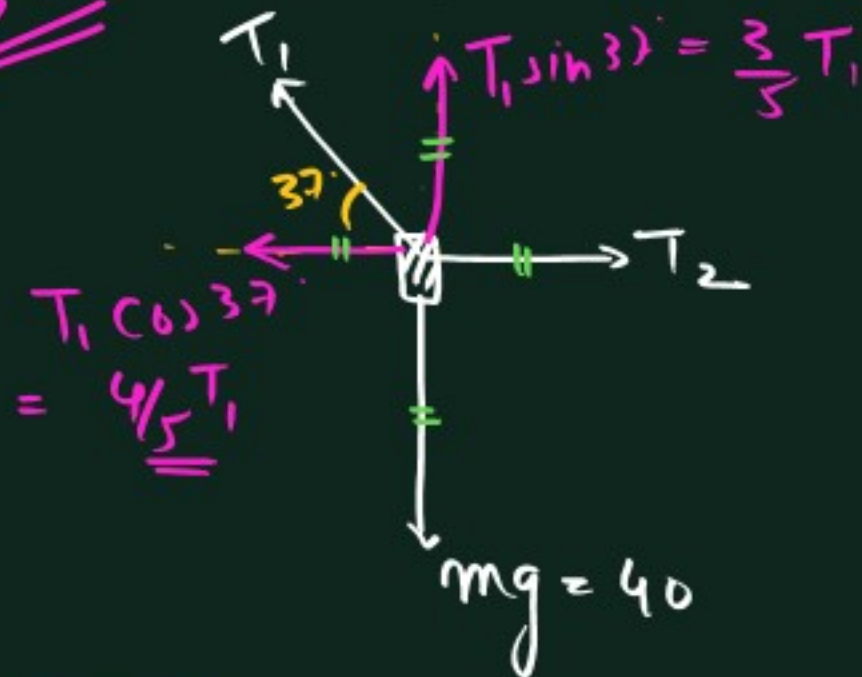


$$(g = 10 \text{ m/s}^2)$$

Find T_1 & T_2 .



Step I



$$\Sigma F_x = 0 \Rightarrow T_2 - \frac{4}{5}T_1 = 0$$

(or)

Force on left = Force on right

$$\Rightarrow \frac{4}{5}T_1 = T_2 \quad \text{--- (1)}$$

$$\Sigma F_y = 0 \Rightarrow \text{Force up} = \text{Force down}$$

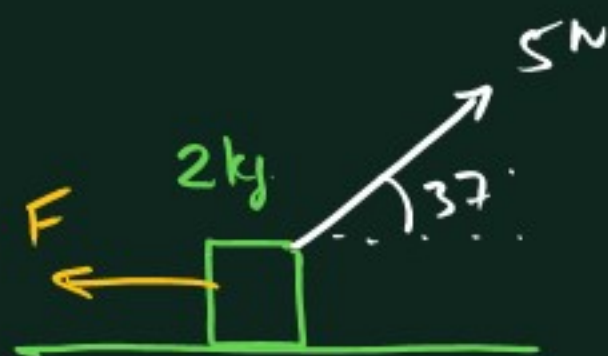
$$\Rightarrow \frac{3}{5}T_1 = 40 \Rightarrow T_1 = \frac{200}{3}$$

Put T_1 in (1)

$$T_2 = \frac{4}{5} \times \frac{200}{3}$$

$$= \frac{160}{3} \text{ N}$$

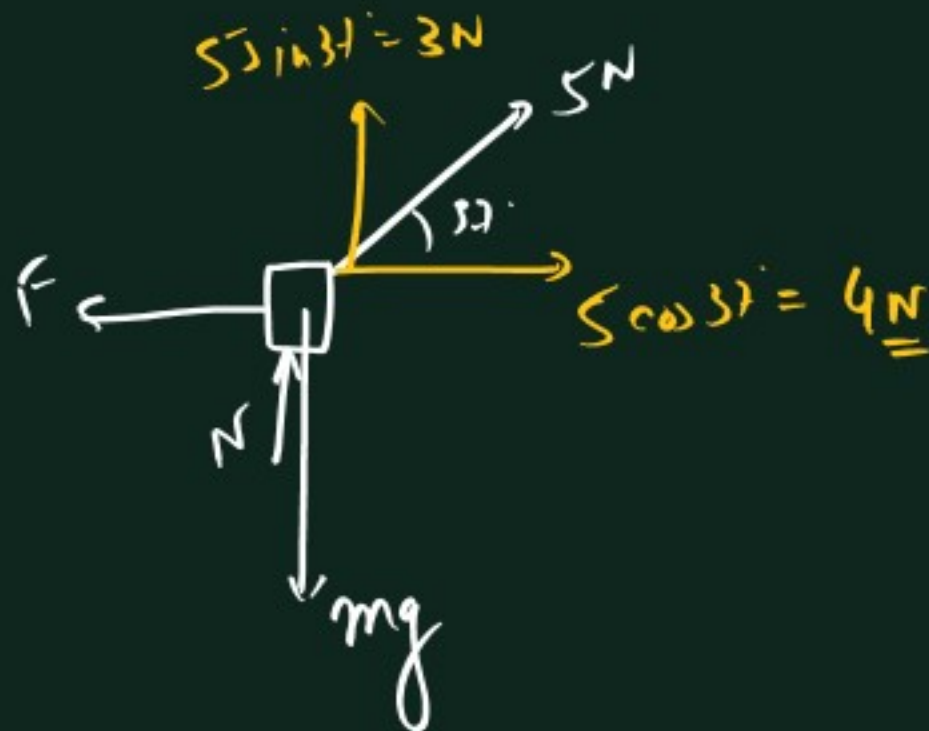
Q2



If box is at rest find

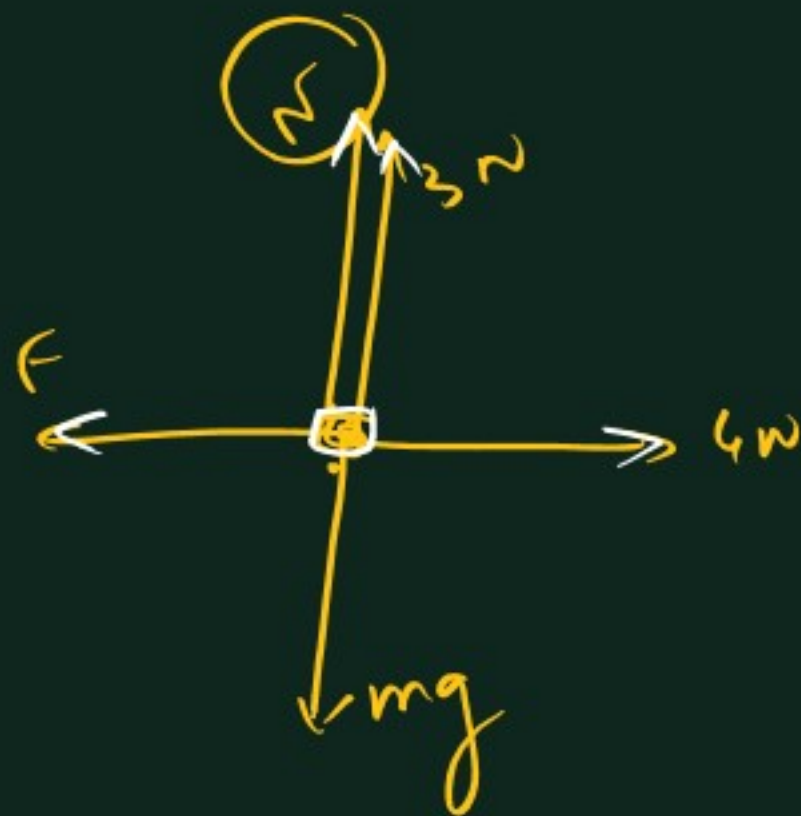
1) $F = ?$

2) Normal force?



$$\underline{\underline{\sum F_x = 0}} \Rightarrow F_{\text{left}} = F_{\text{right}}$$
$$\Rightarrow \boxed{F = 4N}$$

$$\underline{\underline{\sum F_y = 0}} \Rightarrow F_{\text{up}} = F_{\text{down}}$$
$$\Rightarrow 3 + N = \textcircled{mg}^{20}$$
$$\Rightarrow \boxed{N = 17N}$$



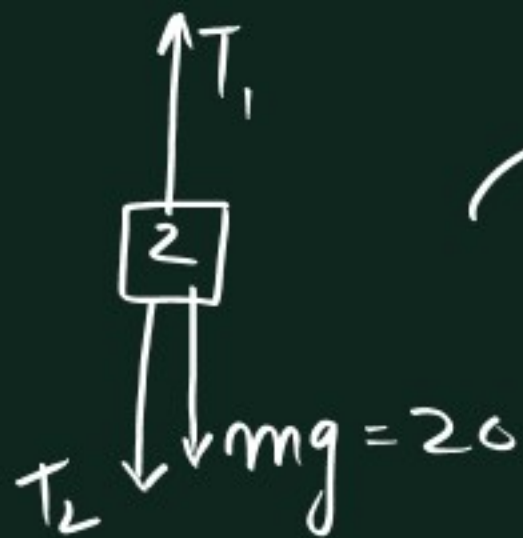
Q3.



1) Draw FBD of both boxes.

2) Find T_1, T_2

($g = 10 \text{ m/s}^2$)



$$\sum F_y = 0$$

$$\Rightarrow T_1 = T_2 + 20 \quad \text{--- (1)}$$



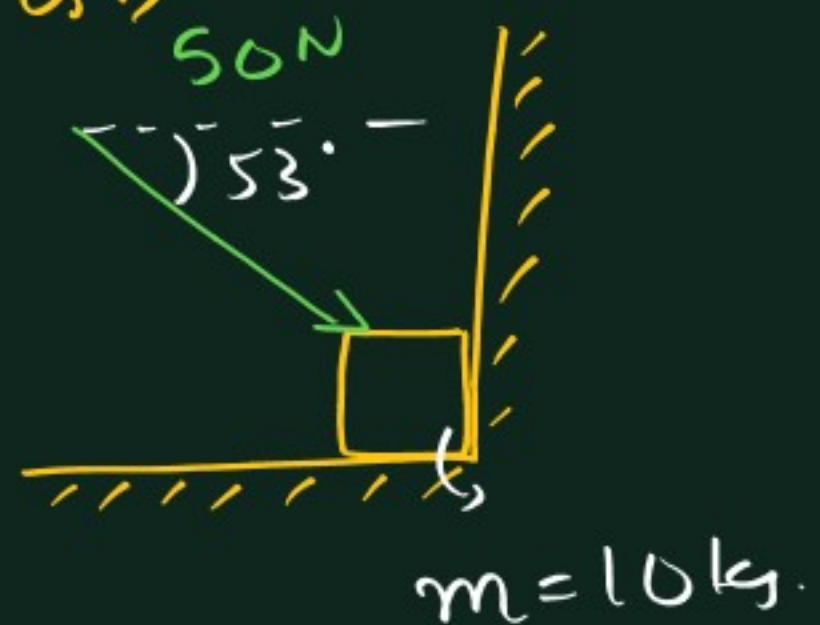
$$\sum F_y = 0$$

$$\Rightarrow T_2 = 50$$

Put in (1)

$$T_1 = 70 \text{ N}$$

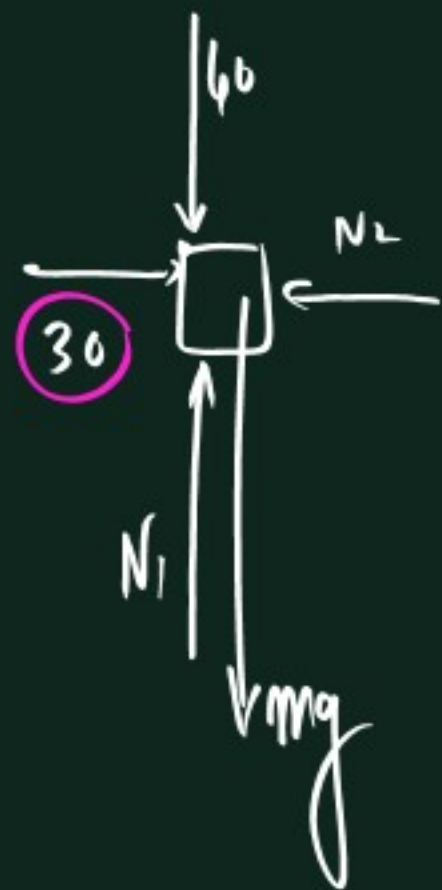
Q4)



1) Draw FBD

2) Find N by wall & ground

$$\left(\begin{array}{l} \cos 53^\circ = 3/5 \\ \sin 53^\circ = 4/5 \end{array} \right)$$



$$\Rightarrow \Sigma F_x = 0 \Rightarrow N_2 = 30$$

$$\Rightarrow \Sigma F_y = 0 \Rightarrow N_1 = 40 + 100$$

$$\Rightarrow N_1 = 140 \text{ N}$$

Q

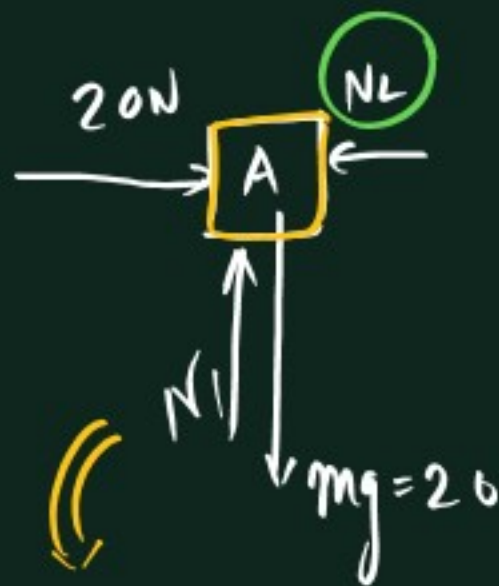


1) Find force b/w A & B
(A $\leftrightarrow$ B) 20 N

2) Find N by ground

m A & B

(20, 40)

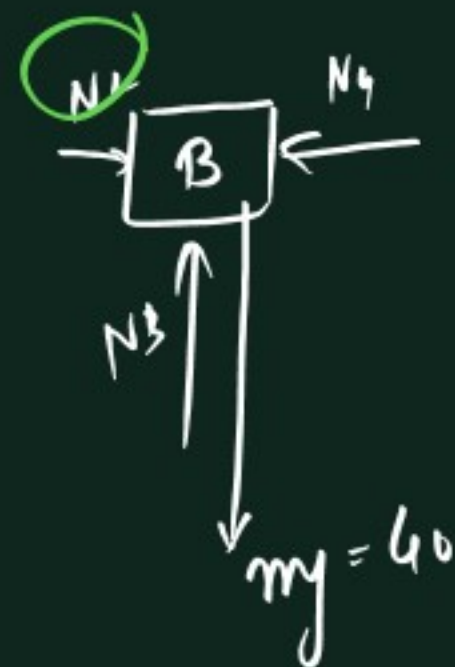


$$\Sigma F_x = 0$$

$$\Rightarrow N_2 = 20 \text{ N}$$

$$\Sigma F_y = 0$$

$$\Rightarrow N_1 = m_A g = 20 \text{ N}$$



$$\Sigma F_x = 0$$

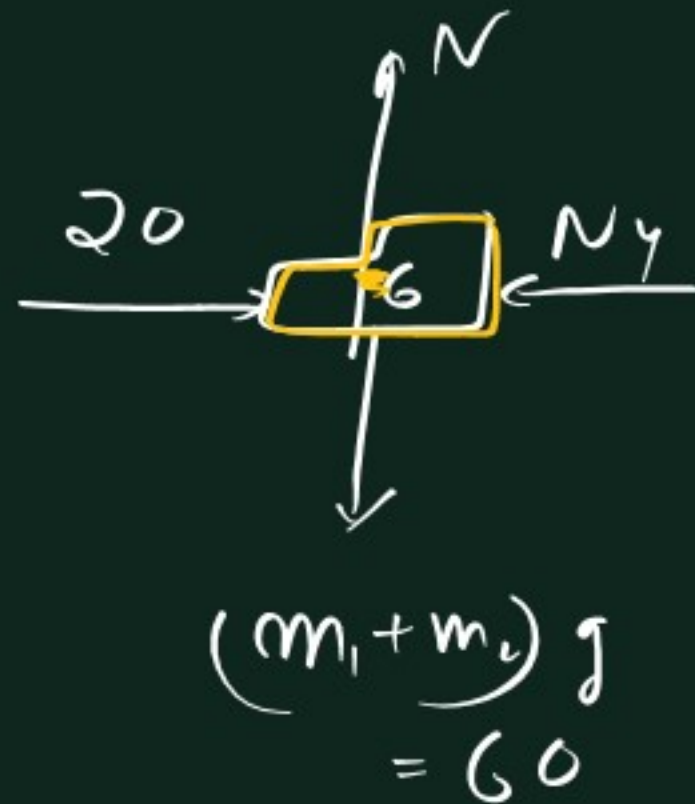
$$\Rightarrow N_2 = N_4$$

$$20$$

$$\Rightarrow N_4 = 20 \text{ N}$$

$$\Sigma F_y = 0$$

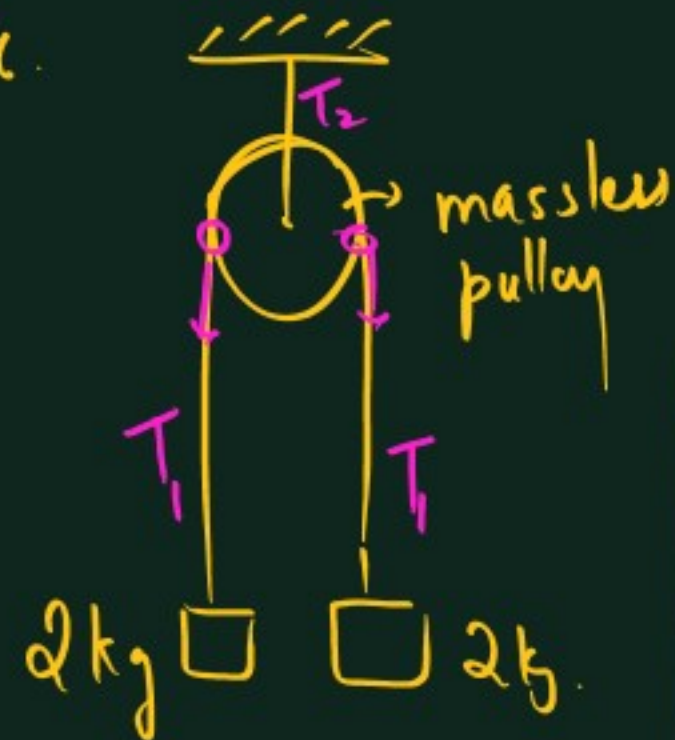
$$\Rightarrow N_3 = m_B g = 40 \text{ N}$$



$$\text{Net } F = m \cdot a^0$$

$$\Rightarrow \text{Net } F = 0$$

Q1.



- 1) Draw FBD of boxes & pulley
- 2) Find T_1 & T_2



$$\Sigma F = 0$$

$$\Rightarrow \boxed{T_1 = 20}$$



Pulley $\Sigma F = 0$

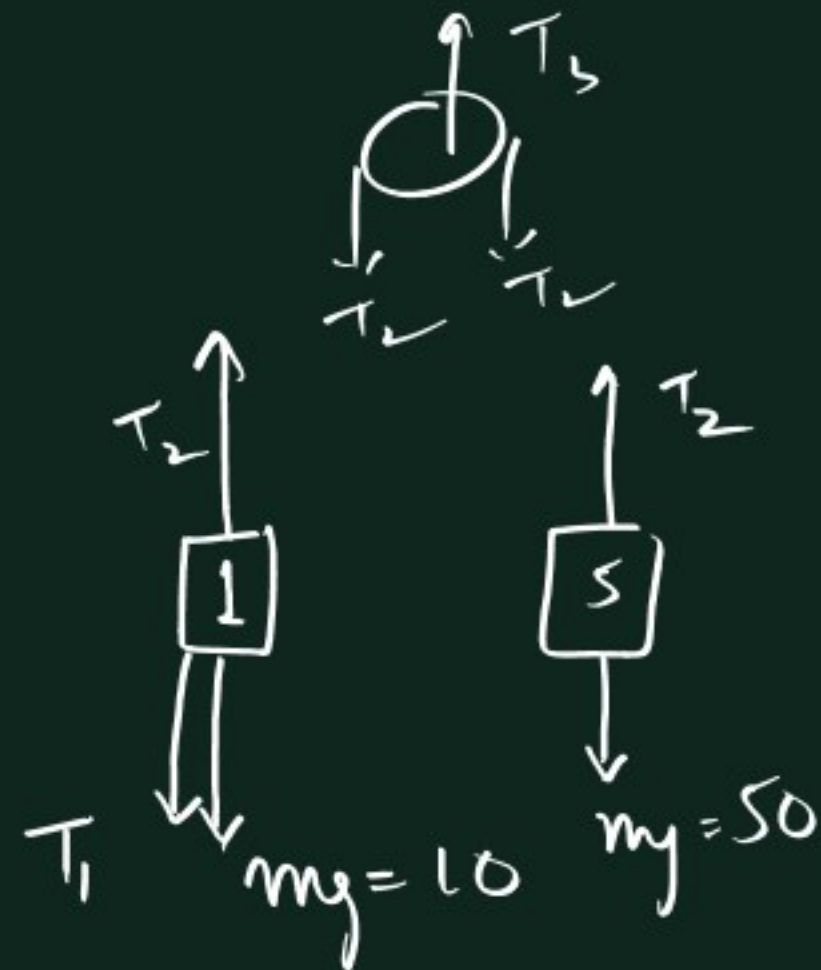
$$\Rightarrow T_2 = 2T_1$$

$$= \underline{\underline{40N}}$$

Q2



- 1) Draw FBD
- 2) T_1, T_2, T_3 !



5kg $\Sigma F = 0$

$$\Rightarrow \underline{\underline{T_2 = 50}}$$

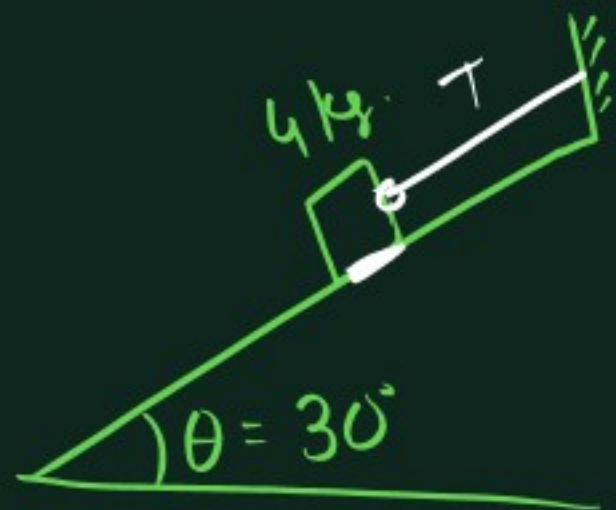
1kg $T_2 = T_1 + 10$

Pulley $\Rightarrow T_1 = T_2 - 10 = \underline{\underline{40N}}$

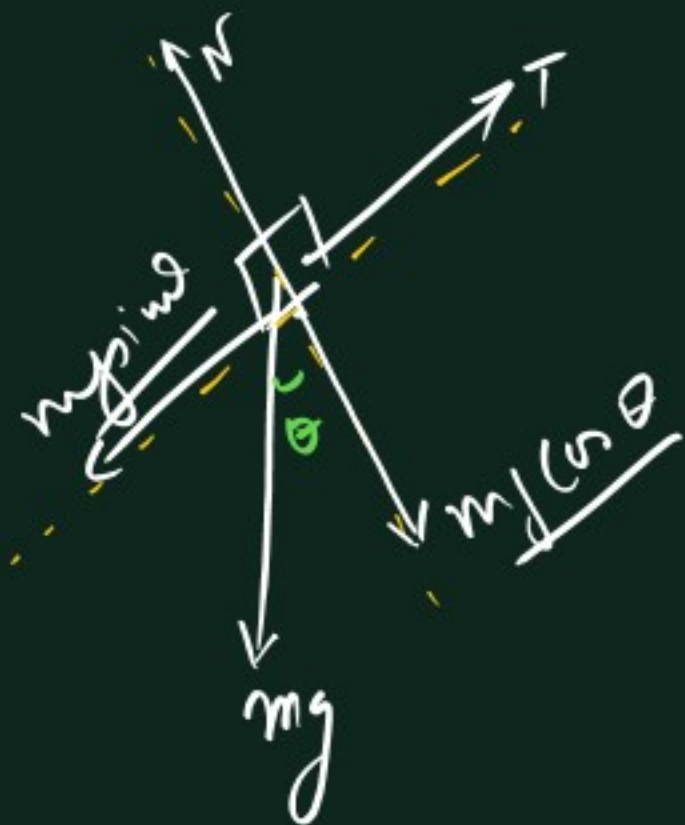
$$T_3 = 2T_2 = \underline{\underline{100N}}$$

Incline plane

Q



To find 1) T in string
2) N on box

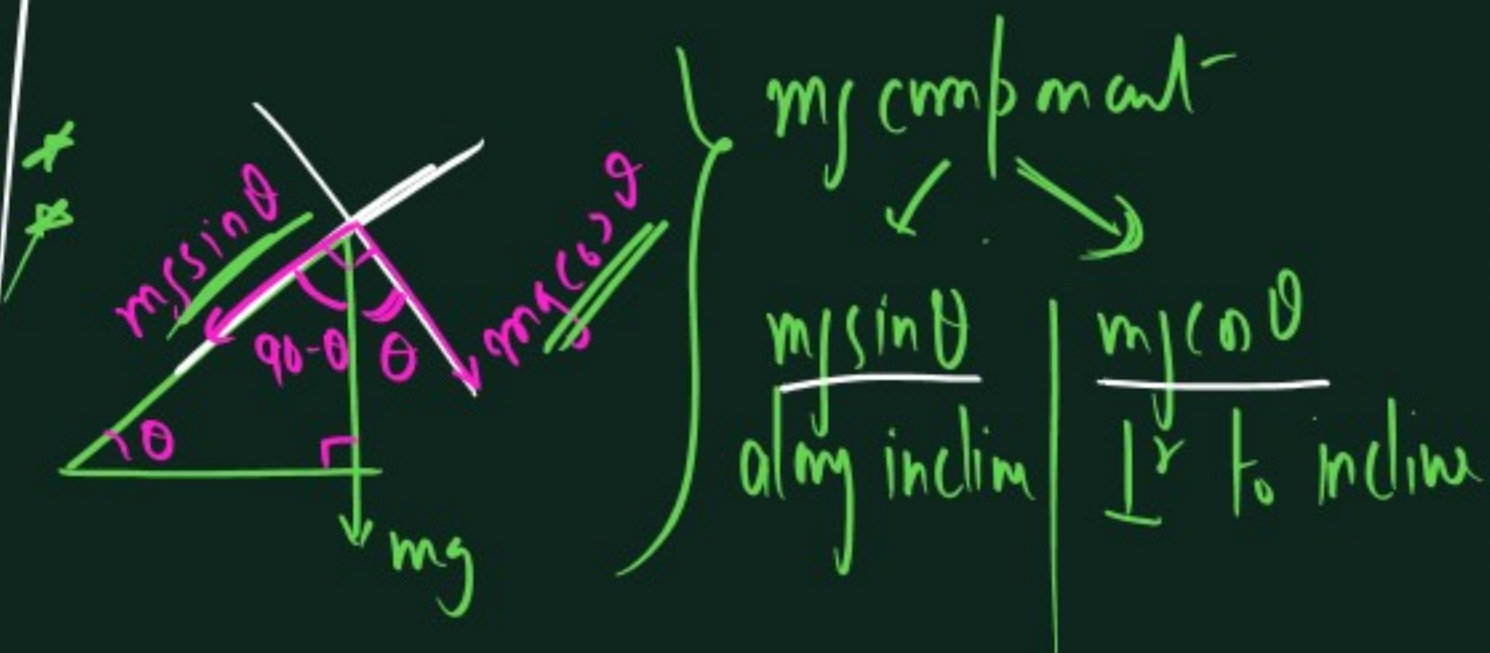


$$\sum F_x = 0$$

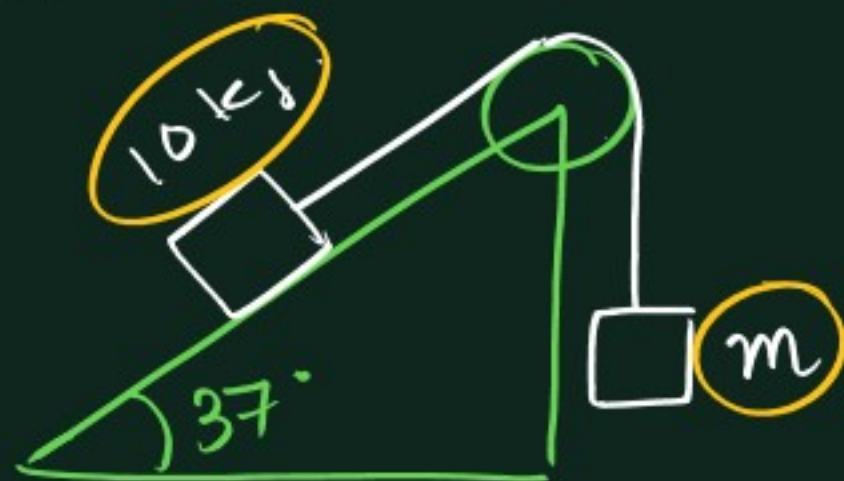
$$\begin{aligned} \Rightarrow T &= mg \sin \theta \\ &= 4 \times 10 \times \frac{1}{2} \\ &= 20 \text{ N} \end{aligned}$$

$$\sum F_y = 0$$

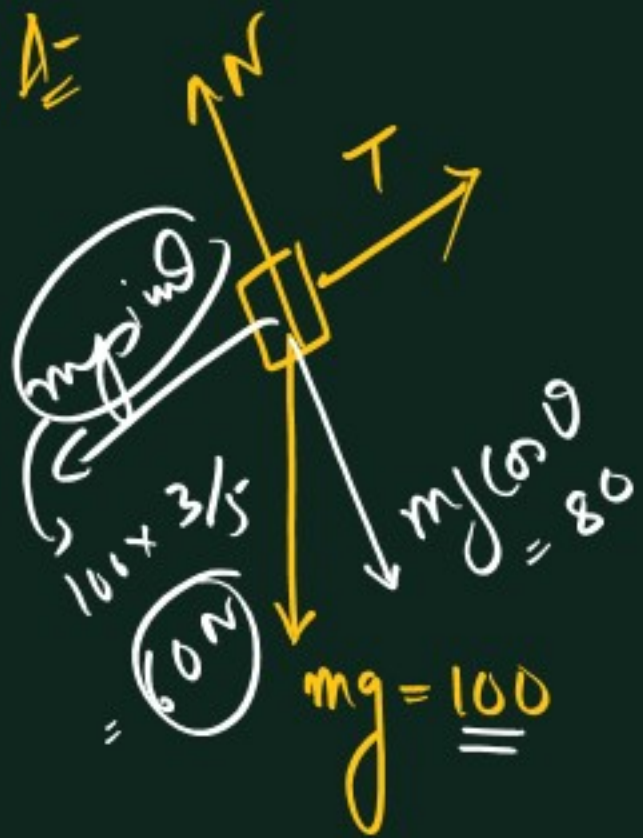
$$\begin{aligned} \Rightarrow N &= mg \cos \theta \\ &= 20\sqrt{3} \end{aligned}$$



Q



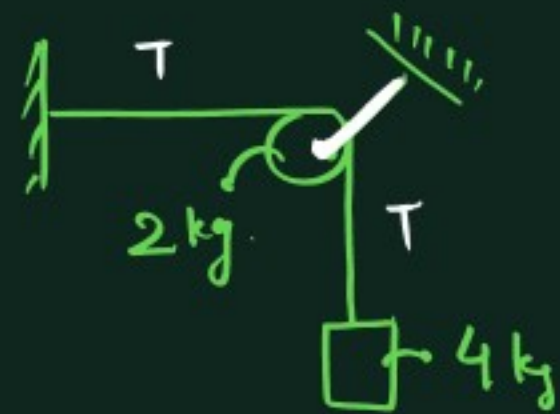
If system at rest, find m .



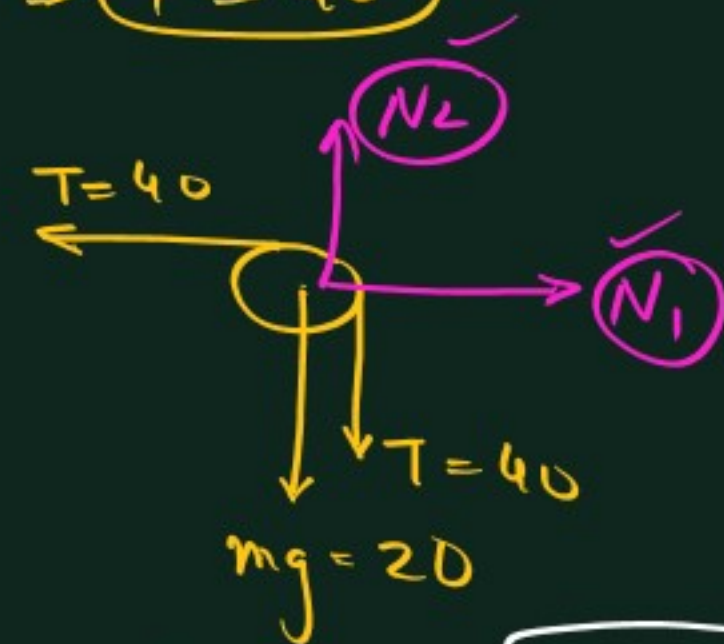
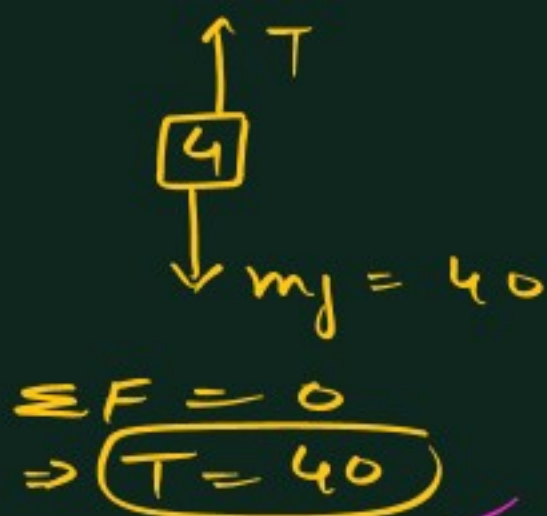
10 kg. $\Sigma F_x = 0$
 $\Rightarrow T = mg \sin \theta$
 $\Rightarrow \boxed{T = 60 \text{ N}}$



$\Sigma F = 0$
 $\Rightarrow T = mg$
 $\Rightarrow m = \frac{T}{g} = \frac{60}{10} \Rightarrow \boxed{m = 6 \text{ kg}}$



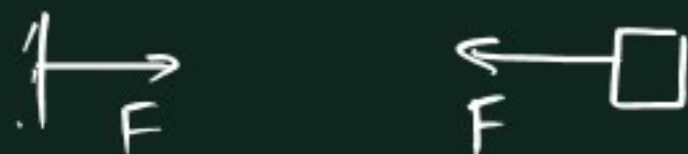
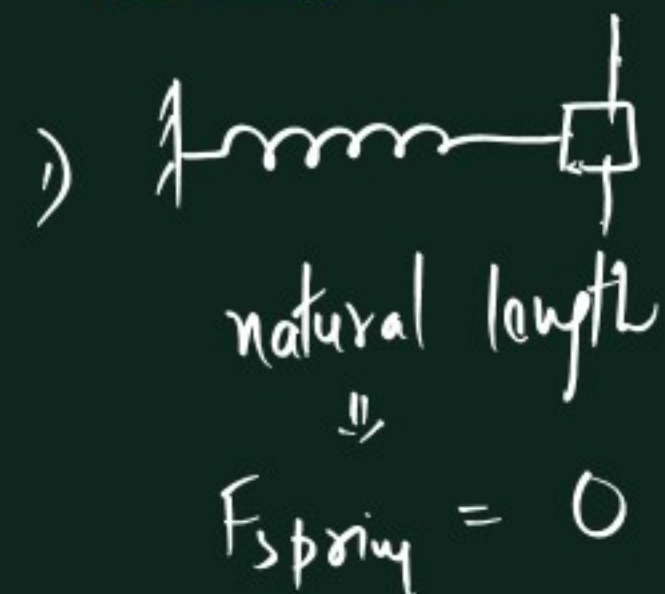
- Q 1) Find T in string
 2) Find force by clamp on pulley



$\Sigma F_x = 0 \Rightarrow \boxed{N_1 = 40}$
 $\Sigma F_y = 0 \Rightarrow \boxed{N_2 = 60}$

Force by clamp
 $= \sqrt{N_1^2 + N_2^2}$
 $= 10\sqrt{52}$
 $= \underline{\underline{20\sqrt{13}}}$

Spring Force



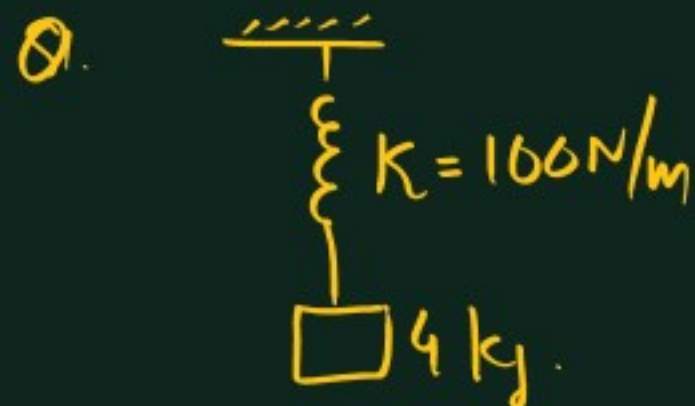
4) Force by spring

$$F \propto x$$

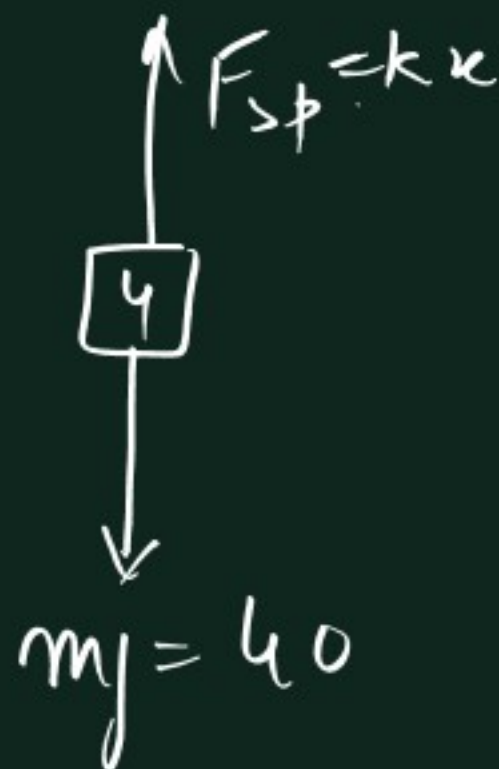
$$|F| = kx$$

k = spring constant (property of spring)
 (unit N/m)

x = elongation/compression
 in spring



Find elongation/
 compression in
 spring?

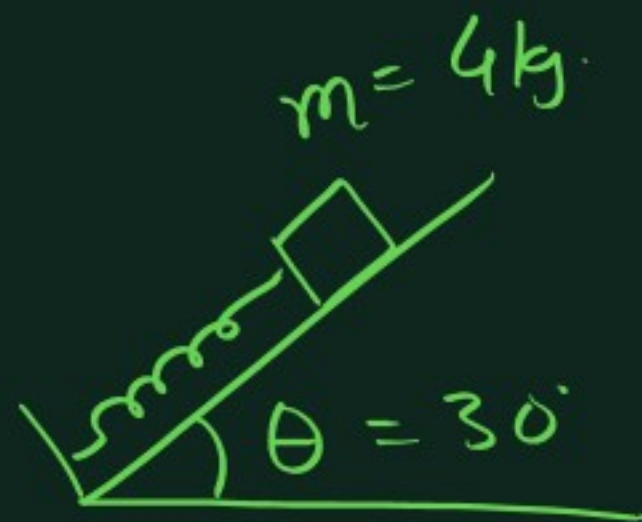


$$kx = m_j$$

$$\Rightarrow x = \frac{m_j}{k}$$

$$\Rightarrow x = \frac{4 \times 10}{100} = \underline{\underline{0.4 \text{ m}}}$$

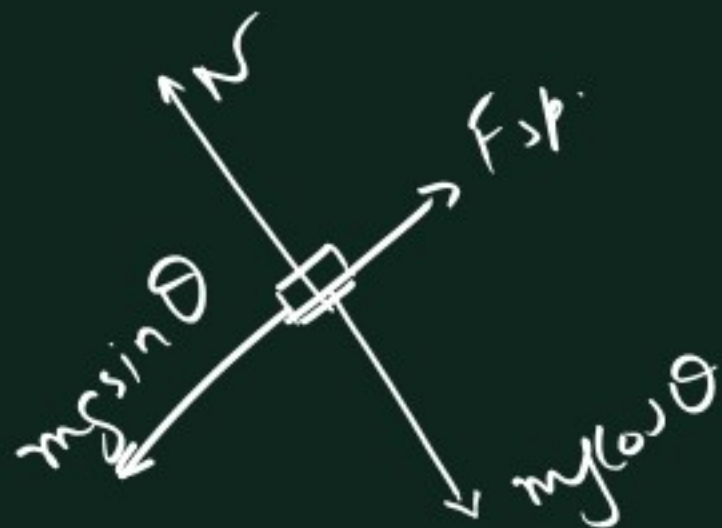
Q2.



$$k = 50 \text{ N/m}$$

Find 1) change in length
of spring

2) N on block.



$$\rightarrow N = mg \cos \theta = \underline{\underline{20\sqrt{3}}}$$

$$\rightarrow F_{sp} = mg \sin \theta$$

$$\Rightarrow kx = 20$$

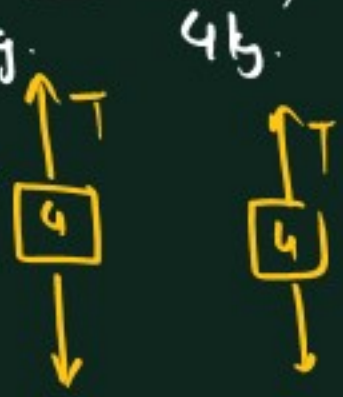
$$\Rightarrow x = \frac{20}{k} = \underline{\underline{0.4 \text{ m}}}$$

$$\text{Reading} = \frac{F_{\text{spring}}}{g}$$

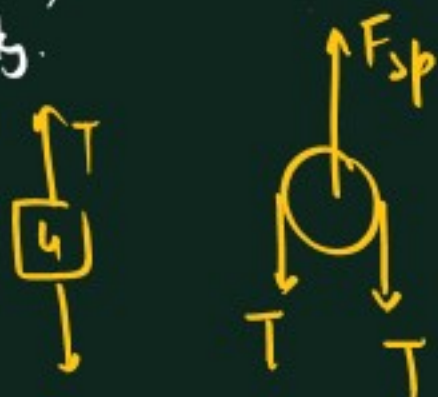
Q3. Find reading of spring balance

1) in Newton

2) in kg



$$mg = 40$$



$$mg = 40$$

$$T = 40$$

$$F_{sp} = 2T = 80N$$

$$\rightarrow \text{Reading (in N)} = 80N$$

$$\rightarrow \text{Reading (in kg)} = 80/g = 8 \text{ kg}$$

Q4.

massless spring balance



System at rest

Find reading of spring balance

1) in Newton

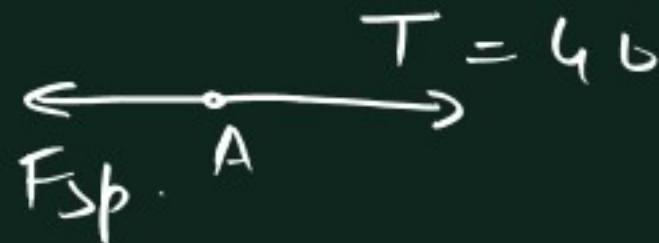
2) in kg



$$T = mg = 40N$$



(A) FBD

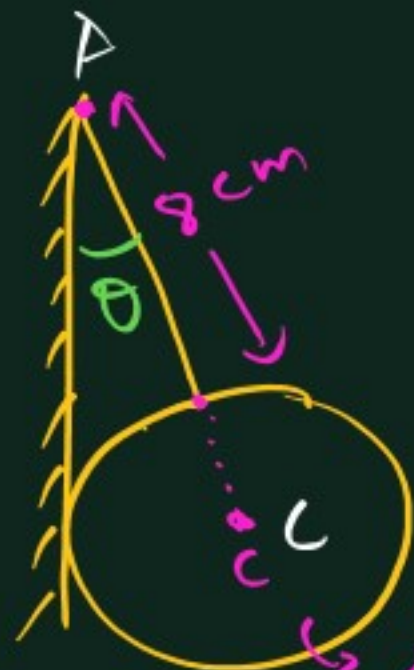


$$\Sigma F = 0$$

$$\Rightarrow F_{sp} = 40N$$

Reading $\rightarrow 40N$
 $\rightarrow 4 \text{ kg}$

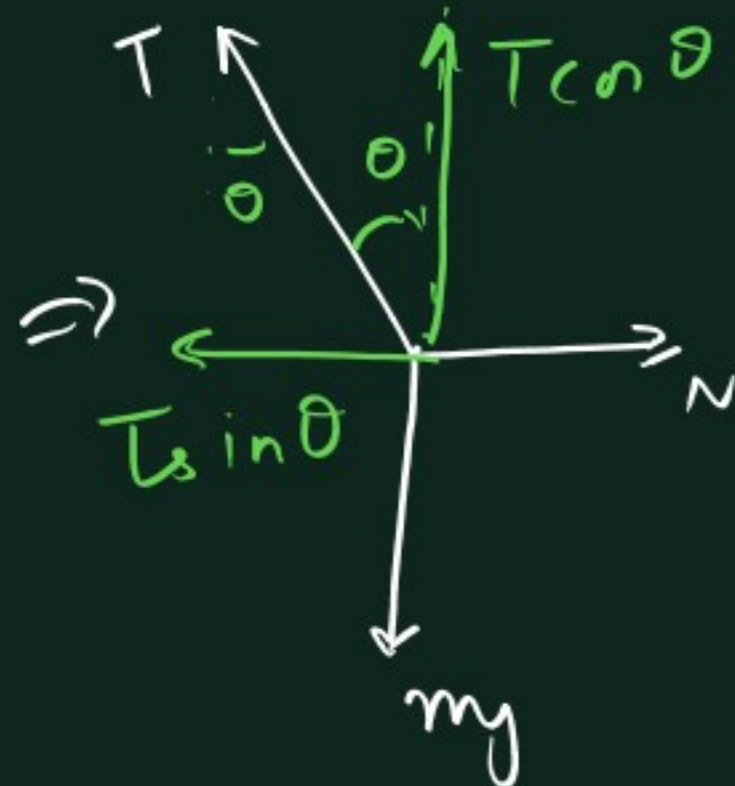
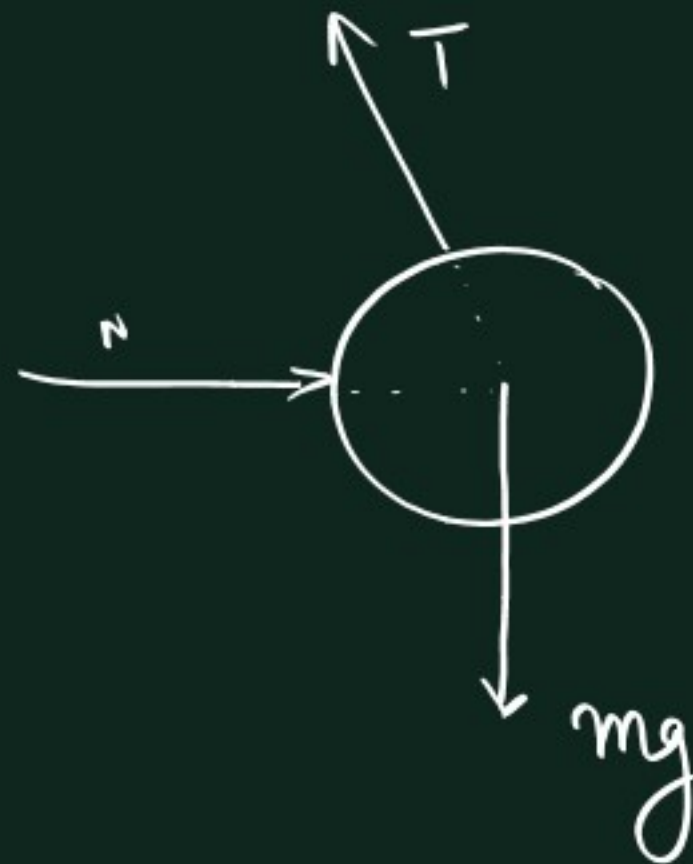
Q



Radius = 5 cm

 $m = 4 \text{ kg}$ Find T in string

$$\cos \theta = \frac{12}{13}$$



$$\sum F_x = 0 \Rightarrow T \sin \theta = N \quad \text{--- (1)}$$

$$\sum F_y = 0 \Rightarrow T \cos \theta = mg$$

$$\Rightarrow T = \frac{mg}{\cos \theta} = \frac{40 \times 13}{12} = \frac{130}{3} \text{ N}$$